# Dumindu Kumarapeli

+94 76 085 8628 | duminduku.25@gmail.com | Colombo, Sri Lanka
linkedin.com/in/dumindu-kumarapeli-7a4636309 | github.com/Dumee-25 | portfolio.kadpkumarapeli.workers.dev

Data Science undergraduate building LLM systems end to end (RAG, agents, local-model inference), with first-author research on privacy-preserving LLM analysis, epidemic forecasting and clinical prediction.

## Education

**NSBM Green University** Sri Lanka
*BSc (Hons) Data Science, currently Year 3* *2025 – Expected Dec 2028*

- GPA 3.85/4.00.

## Research and Publications

- **D. Kumarapeli** and H. Weerasinghe, "Privacy-preserving automated exploratory data analysis using deterministic summarization and local large language models," *Proc. NSBM Student Research Symposium on Innovation Technology (NSRSIT)*, 2026, pp. 146–150. **Best Overall Research**, session award.
- **D. Kumarapeli** and H. Weerasinghe, "Timescale separation and model reduction in operational dengue forecasting," ICCMM 2026, University of Colombo. [accepted, oral presentation]
- Extended NSRSIT paper on DataSense v2: measurable privacy, checked LLM output. Target: ICARC. [in preparation]
- **D. Kumarapeli** and H. Weerasinghe, "A hybrid compartmental and neural network model for district-level dengue forecasting in Sri Lanka: forecast skill, spatial heterogeneity and identifiability." Target: *Ceylon J. Sci.* [in preparation]
- **D. Kumarapeli** and H. Weerasinghe, "Proxy labels versus observed endpoints: an audit and benchmark of clinical benefit prediction on MSK-CHORD." [in preparation]

## LLM and AI Projects

**DataSense v2: Privacy-Preserving Automated EDA** | *Polars, Presidio, OpenDP* portfolio case study

- Rebuilt the NSRSIT system local-first: privacy tiers limit each prompt, Presidio finds personal data, optional differential privacy adds noise, and every number the model writes is checked against the facts it cites.
- Built a benchmark on the frozen engine: it found 4,644 of 5,421 problems planted in 50 OpenML tables, and 27 of the 35 columns it flagged in Raha's real dirty tables had wrong cells. It still misses typos.

**OWL: Datacenter AI Operations Platform** | *Gemini, MCP* github.com/Dumee-25/OWL-Datacenter_agent

- Built an agentic RAG assistant over ChromaDB with hybrid vector and keyword retrieval and recency re-ranking, served through REST, WebSocket and a nine-tool MCP server.
- Paired an Isolation Forest and LSTM autoencoder ensemble with Gemini-generated root-cause reports; every alert carries a confidence score and the system never acts on infrastructure.
- Wrote a simulator producing 4.7M telemetry rows over 18 months; logged per-call LLM latency and cost.

**Fulyn: Private Tool-Calling Agent** | *pgvector, Ollama* github.com/Dumee-25/Fulyn

- Agent turns free-text diary entries into linked expense, sleep, mood and journal records; every tool call is schema-checked and invalid arguments are returned to the model to retry.
- Replaced vector-only search after finding unrelated queries scored 0.62 similarity, higher than some true matches; now merges vector and full-text rankings and flags keyword-free hits.
- Ran opt-in tool-calling tests across local models; llama3.2 failed most by picking the wrong tool or inventing arguments.

## Technical Skills

**LLM / RAG:** Retrieval-augmented generation (RAG), AI agents and tool calling, prompt engineering, Ollama, Gemini, OpenAI API, ChromaDB, pgvector, Model Context Protocol (MCP)
**ML / Data:** PyTorch, scikit-learn, XGBoost, LightGBM, pandas, Polars, NumPy, SciPy, YOLOv8, OpenCV
**Privacy:** Differential privacy (OpenDP), personal-data detection (Microsoft Presidio), prompt-injection filtering
**Languages:** Python, SQL, Java, JavaScript, TypeScript, R
**Backend / Tools:** FastAPI, PostgreSQL, SQLite, Docker, Git/GitHub, Linux, Next.js, React, CUDA
**Spoken:** English (fluent), Sinhala (native)

## Awards

Best Overall Research, session award, NSRSIT 2026 (2026) | Data Storm 7.0 Final Round, Top 10, team Data Drifters (2026)